Name: LIM HONGHWAN

Date of the experiment: 2026/10/13/Th. (synth.)
2026/10/15/Th. (characterisation)

Datasheet for One pot Aldol Condensation and Diels-Alder Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

- Actual mass of benzil used (g) 0.2080
- Actual mass of 1,3-diphenylacetone used (g) fixed. (fwd pipetting).
- Amount of styrene used (mL) 0.4 (bottom miniscus, eye level).

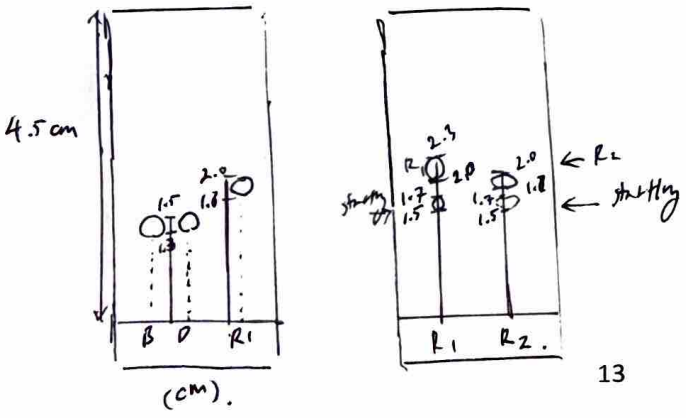
Product

- Mass of empty plastic bag (g) 1.3105 g.
- Mass of plastic bag + product (g) 1.5135 g.
- Mass of your final product (g) 0.2030 g.
- Percentage yield (%): 41.6 ~ 44.1 %.

Observation during the experiment:

start - end.
176.4 – 187.8 °C
started melting, brownish
fully molten, black liquid.

Draw your TLC plate here and label all the distances (in cm)



R2 heating
30 min
↓
TLC 11:42
↓ ~
11:59
confirmed to be finished
bright yellowish brown
(coffe-latte ~ milk tea color)